

Chapter 49. Prone Positioning in Acute Respiratory Failure

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Prone Positioning in Acute Respiratory Failure: Introduction

Approximately 35 years ago, the use of the prone position was proposed to improve arterial oxygenation in patients with acute respiratory failure (ARF).¹⁻³ The prone position, however, may have variable effects on gas exchange. Moreover, it has been suggested that, independent of gas exchange, the prone position may decrease the harm of mechanical ventilation, improving the outcome of patients with ARF. This chapter discusses the mechanisms affecting the changes in gas exchange consequent to the prone position in patients with ARF, and the effects of prone positioning on outcome of critically ill patients.

Effects of Prone Positioning on Gas Exchange

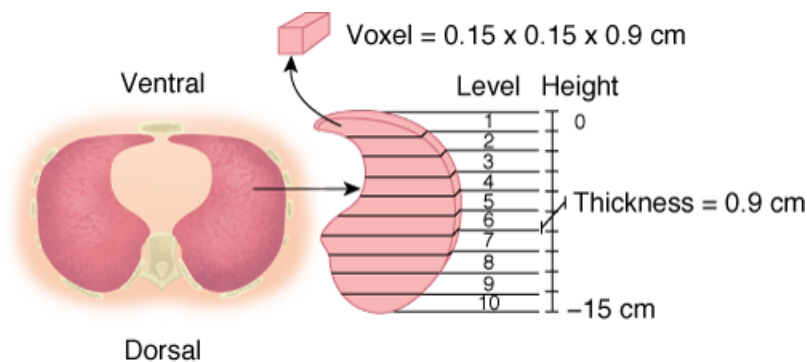
The effects of prone positioning on gas exchange may result from a combination of the following mechanisms: (a) changes in regional lung inflation, (b) redistribution of ventilation, and (c) redistribution of perfusion. These three mechanisms apply to both the normal and diseased lung. In the diseased lung, however, these mechanisms are also affected by the underlying pathology. Moreover, because the underlying pathology is an evolving process, it is likely that the effects of positioning on arterial oxygenation will vary with time.

Regional Lung Inflation

Methods of Investigation

Most of the studies dealing with regional lung inflation in normal subjects were performed with radioactive xenon.^{4,5} We used computed tomography (CT) to quantify regional lung inflation.^{6,7} The CT scan provides a computer-reconstructed image that is composed of several hundred elementary units (voxels). Each voxel is characterized by a given level of absorption of X-rays, which mainly reflects the density of the material being studied. The density is usually expressed in CT numbers or Hounsfield units (H).⁸ A density equal to 0 H characterizes a voxel composed of water, while a voxel with a density of -1000 H is composed of gas. A voxel with a CT number equal to -500 H has a composition of 50% gas and 50% tissue. By analysis of the CT numbers, we quantitatively describe the regional lung inflation of a single CT section, at the level of the lung base, that is representative of the entire lung.⁶ The CT section is divided into ten levels along the vertical axis, each level including approximately 300 to 400 voxels (Fig. 49-1). The gas-to-tissue ratio, which is our index of regional lung inflation, was computed from the average CT number at each lung level.

Figure 49-1



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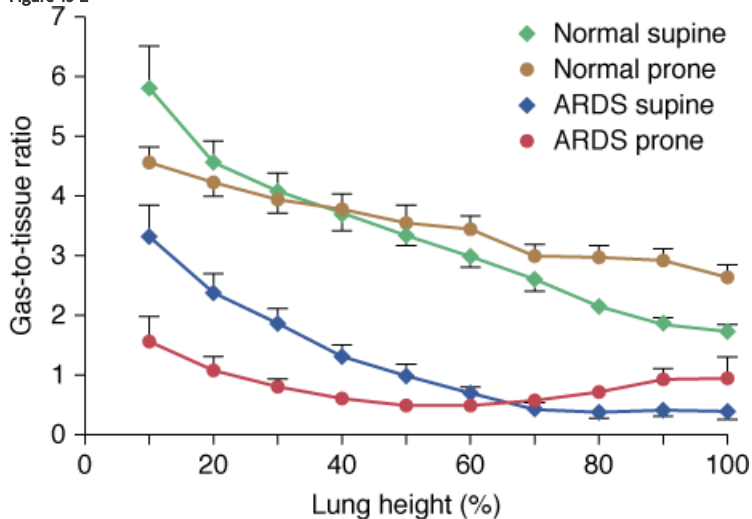
Regional analysis of the CT scan image. The vertical distance from ventral to dorsal surface (height) is divided into ten equal intervals. Ten lung levels are then obtained. Each level is composed of 300 to 400 elementary units (voxels), each characterized by a given CT number expressed in Hounsfield units (H). (Used, with permission, from Pelosi et al.¹⁶)

Normal Lungs

Regional Inflation in the Supine Position

As shown in Figure 49-2, the regional inflation, expressed as the gas-to-tissue ratio, decreases along the vertical axis (from ventral to dorsal). Because the decrease is exponential, the rate of change of regional lung inflation with height may be characterized by a decay constant (K_d). The lower the value of the K_d , the higher the rate of the decrease of regional lung inflation along the vertical axis. The K_d in normal subjects is 13.6 ± 2.5 cm. At this distance from the ventral surface, the gas-to-tissue ratio is 37% of the gas-to-tissue ratio value computed at the ventral surface. In other words, the alveolar dimensions, in the dorsal regions, should be approximately one-third of those at the ventral surface.

Figure 49-2



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The gas-to-tissue ratio as a function of lung height in the supine (diamonds) and prone (circles) positions. The green and brown symbols refer to normal lungs ($N = 14$) and the blue and red symbols refer to lungs of patients with acute respiratory distress syndrome (ARDS) ($N = 20$). It is important to note that "height = 0" refers to the ventral surface in the supine position and to the dorsal surface in the prone position. The height scale direction is from nondependent (0) to dependent (100).

Regional Inflation in the Prone Position

When a normal subject is shifted from supine to prone, the regional inflation distribution changes, increasing in the dorsal regions and decreasing in the ventral regions (see Fig. 49-2). The regional inflation decreases exponentially from dorsal to ventral for both supine and

prone positions; however, the K_d is higher in the prone position compared with the supine position (K_d prone, 26.2 ± 2.2 cm; K_d supine, 13.6 ± 2.5 cm). This indicates that the regional inflation distribution is more homogeneous in the prone than in the supine position, and therefore the prone position does not simply reverse the regional inflation distribution. We confirmed these concepts in animal studies by means of CT scan.⁹

Mechanism of Vertical Inflation Gradient

The regional inflation is believed to depend on the local transpulmonary pressure (i.e., the absolute difference between pressure in the alveoli and pressure at the pleural surface [Ppl]). At functional residual capacity, alveolar pressure equals atmospheric pressure; thus, changes in transpulmonary pressure are caused by regional differences in Ppl. The Ppl gradient in normal subjects is approximately 0.2 to 0.3 cm H₂O/cm.¹⁰ The nature of this pressure gradient, however, is still controversial and is mainly attributed to the following factors: (a) lung weight, (b) shape and mechanical properties of the chest wall, and (c) shape and mechanical properties of the lung.

The role of lung weight in normal subjects is considered a predominant mechanism by some investigators.^{11–14} It was first stressed by Krueger et al,¹⁵ who found that the normal Ppl gradient was very similar to the normal density of the lung (0.22 g/L). The lung-weight theory assumes that the lung behaves as a fluid and that the hydrostatic pressures are transmitted through the lung parenchyma as in a liquid (i.e., at a given lung height, the superimposed pressure [SP_L] is equal to the product of the height of the lung column and density).

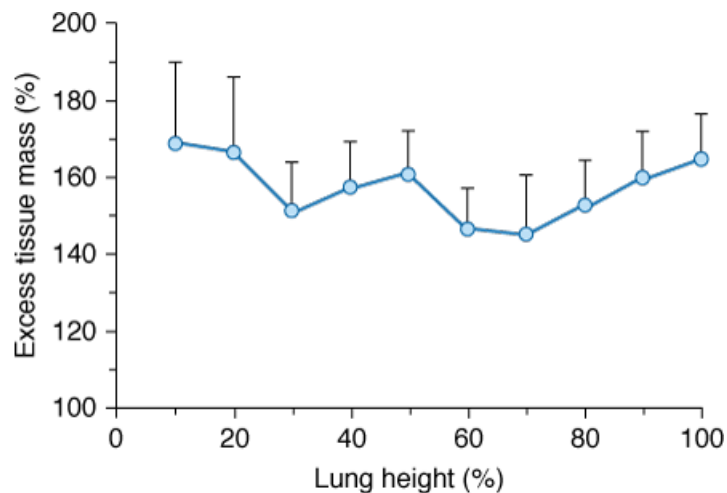
Estimating the SP_L with the CT scan, in normal subjects, we found an average SP_L gradient of 0.23 cm H₂O/cm,¹⁶ which is similar to the Ppl gradient classically reported in physiology. Factors other than lung weight, however, may play a substantial role in determining Ppl. In fact, in the prone position, the decay of the regional inflation with height is less than in the supine position (i.e., the Ppl gradient is probably lower in the prone than in the supine position; see Fig. 49-2). If the SP_L was the only determinant of Ppl, the decay of the regional lung inflation would be similar in prone and supine positions. As shown in Figure 49-2, however, the regional inflation distribution, along the vertical axis, was different in the supine and prone positions. Our findings are in keeping with those of Mutoh et al¹⁷ in normal dogs, in which the Ppl gradient was directly measured. Factors such as the thoracic lung shape and compliance should then be involved in determining the regional inflation changes with positioning. In addition, the prone position eliminates the compression of the lungs by the heart, which also contributes to a more homogeneous distribution of lung inflation.^{18,19}

Lungs in Acute Respiratory Failure

Anatomic Changes during Acute Respiratory Failure

The early phases of ARF are characterized by lung edema. The edema causes a large increase in “lung tissue” and therefore in lung weight. In ARF patients, we found that lung weight, as measured by the CT scan technique, was two to three times greater than normal,²⁰ and this is consistent with the lung weight values reported in autopsy series.²¹ We also found that edema (i.e., the amount of tissue mass in excess of normal [excess tissue mass]) does not accumulate preferentially in the dependent regions, and it is quite evenly distributed at each lung level (Fig. 49-3). The nongravitational distribution of edema was reported in a series of animal experiments using different techniques.^{22,23} This homogeneous increase in lung edema has two important consequences for regional inflation. First, although ARF is characterized by a marked decrease in lung gas volume,²⁰ the total lung dimensions are similar to normal secondary to the increase in lung tissue mass (edema).¹⁶ Second, the increased lung tissue mass leads to a dramatic increase in SP_L, which may be as high as 10 to 15 cm H₂O in the most dependent regions.^{6,7,16}

Figure 49-3



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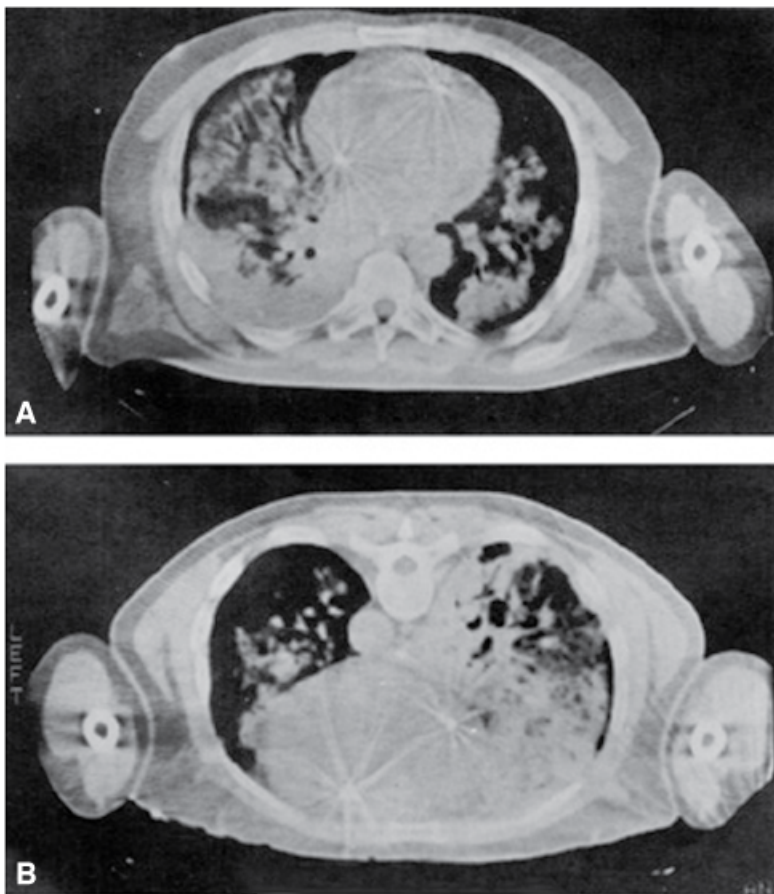
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The distribution of excess tissue mass (i.e., the mass in excess of the expected normal mass) as a function of lung height. Data refer to thirty-four ARF lungs studied in the supine position. Zero on the abscissa refers to the ventral lung surface. (Used, with permission, from Gattinoni et al.⁵¹)

Regional Inflation in the Supine Position

Figure 49-2 shows the behavior of regional inflation in patients with ARF in the supine position. As in the normal lung, the regional inflation decreases exponentially along the ventral–dorsal axis.¹⁶ Two important differences, however, must be stressed. First, the regional inflation at the ventral surface is only half that of the normal lung. Second, the decay constant in the regional inflation with height (K_D) is almost half that of normal; that is, the rate of the decrease of regional inflation from the ventral to dorsal region is double that of the normal lung (K_D ARF = 6.9 ± 1.2 cm vs. K_D normal = 13.6 ± 2.5 cm). This suggests that in supine ARF patients at 0 cm H_2O positive end-expiratory pressure (PEEP), there is almost total alveolar collapse in the posterior half of their lungs, as confirmed by the CT scan image (Fig. 49-4).

Figure 49-4



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A representative CT scan image in supine (A) and in prone (B) positions. Note the typical redistribution of lung densities. Images taken 10 minutes apart, and at end-expiration apnea, with 10 cm H₂O PEEP. (Used, with permission, from Gattinoni et al.⁶)

Regional Inflation in the Prone Position

Figure 49-2 shows the behavior of regional inflation in the prone position. As with normal subjects, the inflation gradient is reversed, and the regional inflation is greater in dorsal regions and lower in ventral regions. Once again, the dependent lung regions (ventral) are collapsed (see Fig. 49-4). As with normal subjects, regional inflation is more homogeneously distributed than in the supine position; however, exponential fitting of the experimental data did not reach a level of significance in all patients.

Mechanism of Vertical Inflation Gradient

The main difference in regional inflation between normal subjects and patients with ARF is the greater rate of decrease of regional inflation along the vertical axis in ARF lungs. This suggests that the Ppl gradient is higher in patients with ARF than in normal individuals.¹⁶ The most likely explanation of the increased Ppl in ARF is increased lung weight; the overall lung shape, in fact, is not different between normal individuals and patients with ARF.¹⁶ The thoracic-lung shape modifications when the patients are turned prone, however, probably change the Ppl gradient (i.e., the modifications of the lung and the chest wall shape may redistribute the forces acting at the pleural surface).

The relationships between lung shape modifications and distribution of regional lung inflation in ARF patients is discussed in “[Mechanisms Involved in Arterial Oxygenation Changes in the Prone Position](#)” below.

Summary

In patients with ARF, lung densities redistribute from the dorsal to the ventral region when turned prone. Regional inflation decreases exponentially from the nondependent to the dependent lung region in both the supine and prone positions. The decay in regional

inflation, both in normal individuals and in patients with ARF, is greater in the supine than in the prone position. This indicates that regional inflation is usually more homogeneously distributed in the prone position. The regional inflation gradient is secondary to the Ppl gradient. Lung weight is probably a major determinant in increasing the Ppl gradient in ARF. Thoracic–lung shape modifications from supine to prone, however, may produce a more homogeneous distribution of regional inflation in the prone position.

Distribution of Ventilation

As a general rule, ventilation relates to applied pressures, flows, and resistances according to the following equation:

$$DV = C \times (DP_{tp} - \dot{V} \times R)$$

where DV is variation in volume, C is compliance, DP_{tp} is transpulmonary pressure, $\dot{V}$ is flow, and R is resistance. It follows that regional ventilation depends on regional compliances, regional transpulmonary pressure swings, regional flow rates, and regional resistances.

Normal Lungs

Distribution of Ventilation in the Supine Position

When a bolus of radioactive gas is inhaled by spontaneously breathing subjects in the supine position, ventilation is predominantly distributed to the dependent lung regions.²⁴ From available data, the major determinants of the distribution of ventilation appear to be: (a) the regional inflation distribution and (b) the pattern of diaphragmatic movement (active or passive).

Mead et al,²⁵ Otis et al,²⁶ and Pedley et al²⁷ proposed a lung model divided into two compartments: upper and lower. Regional inflation is greater in the upper (ventral) compartment and smaller in the lower (dorsal) compartment. The upper lung regions are closer to the flat portion of their pressure–volume curve, while the lower regions are on the steeper portion of the curve. Consequently, for a given applied pressure, ventilation is greater in the lower lung regions than in the upper regions.

In the supine position, a large portion of the lung is close to the diaphragm, which may vary its tension, shape, and position. During tidal spontaneous breathing, there is a greater displacement of the dependent hemidiaphragm, with consequent greater ventilation of the dependent lung.^{28–30} Accordingly, in spontaneously breathing subjects, both regional inflation and diaphragmatic movement cooperate to distribute ventilation preferentially to the dependent lung. There are, however, three major conditions in which the ventilation is greater in the nondependent lung: (a) ventilation at low lung volume, (b) ventilation at high inspiratory flow rate, and (c) mechanical ventilation during anesthesia and muscle paralysis. The preferential ventilation of the upper compartment at a low lung volume is easily explained by the work of Otis et al²⁶ (i.e., at a low lung volume, the upper lung regions are probably shifted on the steeper part of the pressure–volume curve). At a high inspiratory flow rate, the accessory respiratory muscles appear to play a determining role in the redistribution of ventilation to the upper compartment.^{31,32} During mechanical ventilation, particularly during anesthesia and paralysis, the diaphragm appears to play a major role in the distribution of ventilation.^{33–35} In this situation, the diaphragm behaves as a flaccid membrane, which is faced with the vertical pressure gradient of the abdominal contents. The upper, nondependent part of the diaphragm moves passively and faces a lower abdominal pressure; thus, ventilation is greater in the nondependent lung regions.

Distribution of Ventilation in the Prone Position

Distribution of ventilation in prone position is still controversial. Indeed, some authors found a uniform vertical distribution of ventilation in spontaneously breathing subjects,³⁶ whereas others found a vertical gradient with either greater ventilation in the dependent (ventral) regions⁴ or in the nondependent lung regions.³⁷

No data are available concerning ventilation at low lung volumes or at a high inspiratory flow rate in the prone position. During mechanical ventilation in anesthetized paralyzed normal individuals in the prone position, Rehder et al³⁷ found that the distribution of ventilation was greater in the nondependent lung regions, where the passive upper position of the diaphragm faced a lower abdominal pressure.

Recently, in an animal study, we confirmed that the distribution of ventilation in the prone position is more homogeneous than in the supine position, and that in the prone position there is less lung strain once tidal volume is delivered.⁹

Lungs in Acute Respiratory Failure

Distribution of Ventilation in the Supine Position

CT scan studies in patients with ARF have shown that ventilation is distributed preferentially to the nondependent lung because the dependent lung is usually collapsed and/or consolidated (see Fig. 49-4). We obtained CT scans at both end-expiration and end-inspiration in ten patients during the early phase of ARF.³⁸ In these patients, we, indeed, found that “ventilation” was preferentially distributed to the nondependent (ventral) lung regions. The pattern was in part modified by application of PEEP. With an increase in PEEP and reopening of the dependent lung regions, “ventilation” was more uniformly distributed (gas could reach the recruited dependent regions).

Distribution of Ventilation in the Prone Position

Few data have been published on the distribution of ventilation in patients with ARF placed in the prone position. Because regional inflation is more uniform in the prone position, similarly to the pattern in normal lungs, however, one would expect that ventilation should be more uniform in this posture. Vieillard-Baron et al showed that the prone position does improve homogenization of tidal ventilation by reducing time-constant inequalities.³⁹

Summary

In spontaneously breathing normal subjects, when supine, ventilation is preferentially distributed to the dependent lung regions, although prone position data are controversial. During mechanical ventilation, this pattern is reversed with preferential distribution of ventilation to the nondependent lung in both supine and prone positions. Accordingly, the main determinant of ventilation distribution is the regional inflation gradient and transdiaphragmatic pressures. In mechanically ventilated patients with ARF at low PEEP level, ventilation is preferentially distributed to the nondependent lung regions. In the prone position, we expect a similar pattern of ventilation distribution, because the collapsed areas move from the dorsal to the ventral regions. Tidal volume, however, is more homogeneously distributed in the prone position, because of a reduction in the slow compartment zones of the lung, and possibly because of a more homogeneous distribution of regional inflation.

Distribution of Perfusion

Normal Lungs

Distribution of Perfusion in the Supine Position

The most widely accepted model that describes the pulmonary perfusion pattern is that described by West et al.⁴⁰ According to these investigators, the relationship between blood flow and pulmonary artery pressure, alveolar pressure, and venous pressure can be modeled as in a Starling resistor, where a collapsible tube traverses a closed chamber (the gas compartment) in which pressure (the alveolar pressure) may be varied at will. When the inflow pressure (pulmonary artery pressure) is lower than the chamber pressure (alveolar pressure), blood flow stops (zone 1). When the inflow pressure is higher than the chamber pressure, the flow through the system is governed by either the difference between pulmonary artery pressure and alveolar pressure (zone 2) or the difference between pulmonary artery pressure and venous pressure, when the outflow pressure (venous pressure) exceeds the chamber pressure (zone 3).

According to this “gravitational” view, perfusion should progressively increase down the lung. In the supine position, the lung should be characterized by perfusion in zone 2 and zone 3, increasing from ventral (zone 2) to dorsal (zone 3), because zone 1 conditions do not occur. Several authors, suggesting that factors other than gravity can determine the perfusion heterogeneity, have challenged the model of West et al.^{41–45}

Distribution of Perfusion in the Prone Position

According to the gravitational theory, a perfusion gradient should exist from dorsal to ventral. Few data confirm this pattern in human subjects.⁴ A perfusion gradient, however, could not be observed in the prone position in dogs;^{44,46} regional pulmonary blood flows were positively correlated when measured in supine and prone position (and not negatively correlated, as expected according to the gravitational theory).

Lungs in Acute Respiratory Failure

Distribution of Perfusion in the Supine Position

Whatever the determinants of perfusion in the normal lung, several physiologic and anatomic factors are likely to alter the perfusion distribution in ARF. Among these are (a) hypoxic vasoconstriction, (b) vessel obliteration, and (c) extrinsic vessel compression.

The effects of hypoxic vasoconstriction were evaluated in the lateral position using separate lung ventilation.⁴⁷ The inspiration of a hypoxic gas mixture by the dependent lung failed to redirect blood flow toward the nondependent lung, whereas diversion occurred in the supine position. This suggests that hypoxic vasoconstriction alone is not able to counteract the effects of posture on pulmonary blood flow.

Both macrothrombi and microthrombi have been found at various stages of ARF.⁴⁸ These anatomic alterations were associated with the appearance of filling defects when selective angiography was performed.⁴⁹ Unfortunately, data are not available regarding the distribution of these defects (homogeneous, dependent, or nondependent?), but obviously the presence of vessel obliteration will alter perfusion distribution. Nonthrombotic obliterative vascular disease may also occur in the early stage of ARF secondary to congestion or compression of vessels, and in the intermediate or late stages of ARF secondary to focal obstruction of small arteries and veins as a result of fibrosis.

Increased SP_L in ARF could also cause extrinsic vessel compression: The radiologic equivalent would be the “pruning” of vessels on selective angiography. In early ARF, we found (unpublished observation) that selective angiography was usually positive (filling defects and/or pruning) when the tip of the catheter was positioned in the dependent lung regions (dense areas on CT scan), and negative when it was in the nondependent lung regions. Moreover, we occasionally found changes in the selective angiography findings (from positive to negative) when the patients were turned from supine to prone, and the angiography was repeated in the same lung region.

All these data suggest that the pulmonary flow distribution is altered in ARF. If the compression mechanism and hypoxic vasoconstriction are operating, both should act mainly in the dependent lung regions (less aeration and higher SP_L), and the blood flow should partly be diverted toward the nondependent regions, as observed in cardiogenic pulmonary edema.⁵⁰ This would partly protect against hypoxemia. In fact, it is not unusual to find 60% to 70% of airless lung (in the dependent regions) with a shunt fraction of 30% to 40%.⁵¹ This indirectly proves that, in the supine position, part of the pulmonary blood flow is redistributed to the nondependent lung.

Distribution of Perfusion in the Prone Position

To our knowledge, few data are available regarding the distribution of pulmonary blood flow in patients with ARF mechanically ventilated in the prone position. Experimental evidence in dogs suggests that perfusion to the dorsal regions is greater than to the ventral regions in the prone position.⁴⁶ This occurrence might lead to a more even distribution of blood flow.⁵² Endogenous nitric oxide plays a role in the regulation of regional pulmonary perfusion. In fact, after the inhibition of nitric oxide synthase by N(G)-monomethyl-L-arginine infusion, nitric oxide synthase messenger RNA expression and nitric oxide production were significantly higher in dorsal region as compared with ventral lung regions. In the supine position, lung perfusion was shifted to ventral regions during nitric oxide synthase inhibition, whereas in the prone position lung perfusion remained unchanged.⁵³ Moreover, acute changes of oxygenation consequent to positioning in patients with acute respiratory distress syndrome (ARDS) do not induce any short-term effect on pulmonary endothelin-1 net clearance or angiotensin-II net formation.⁵⁴ These data suggest that mechanisms other than gravity or hypoxic pulmonary vasoconstriction operate in the prone position to distribute blood flow.

Summary

In the normal lung, blood flow is distributed according to gravity. Factors other than gravity, however, may play a role in governing regional perfusion. These factors seem prevalent in the prone position, where the vertical perfusion gradient may disappear. In the edematous ARF lung, available data suggest that pulmonary blood flow is partly diverted toward the nondependent, more aerated, regions. Vessel compression and hypoxic vasoconstriction may play a role in this pattern of blood flow distribution. In the prone position, during ARF, the nondependent, dorsal regions seem more perfused than the ventral regions. Consequently, in ARF, perfusion seems less dependent on gravity than in normal conditions.

Ventilation–Perfusion Matching

In the past, a number of investigators have used the multiple inert gas technique to study ventilation–perfusion ($\dot{V}_A/\dot{Q}$) relationships in different forms of ARF.^{55–57} The general conclusions of these studies can be summarized as follows: although ventilation and perfusion are usually well matched in ARF lungs, a consistent portion of pulmonary flow traverses airless regions (true shunt), and low $\dot{V}_A/\dot{Q}$ compartments are scarcely represented.

The effect of prone positioning on regional shunt and aeration was studied in an animal model of acute lung injury using $^{13}\text{N}_2$ injection and positron emission tomography.⁵⁸ This study confirmed that supine position is associated with a high shunt fraction in the dorsal region (i.e., high perfusion and poor aeration), whereas prone positioning leads to a decrease of shunt fraction in the dorsal region, without a concomitant impairment of gas exchange in the ventral region. These effects were related to an increase in total gas content and a more uniform distribution of aeration in the prone position, because the increase in aeration in the dorsal region was not offset by loss of aeration in the ventral region. Moreover, in the prone position, perfusion resulted in less gravity dependence, because a substantial fraction of pulmonary blood flow was maintained in the dorsal region. In summary, this study demonstrated that prone positioning restores aeration while it preserves perfusion in the dorsal region of the lung, decreasing the regional and total shunt secondary to an increase in gas-exchanging pulmonary blood flow.

Effect of Prone Positioning on Clinical Outcome of Patients with Acute Respiratory Failure

Effects on Arterial Oxygenation

Over the last 35 years a series of reports appeared in the literature dealing with the use of the prone position as a tool to improve arterial oxygenation in patients with ARF. Over the years, several clinical studies on prone positioning have been conducted, including many retrospective studies,^{1,3,59,60} prospective observational studies,^{6,39,61–76} randomized clinical trials,^{77–81} and meta-analyses.^{82–86}

From these published data some generalizations may be made.

1. There is strong evidence that prone positioning is effective in improving oxygenation of patients with ARF. For this beneficial effect, the use of prone positioning is currently considered in the treatment of patient requiring injurious levels of fractional inspired oxygen concentration (FI_{O_2}) and plateau pressure⁸⁷ or as a “rescue maneuver” for life-threatening hypoxemia. It must be pointed out, however, that (a) the degree of improvement is highly variable (from a few to hundreds of millimeters of mercury change in partial pressure of arterial oxygen (Pa_{O_2})), (b) a small fraction of patients do not show any improvement or even show a deterioration in arterial oxygenation while prone, and (c) the individual response to the prone position is hard to predict.
2. In the patients who respond, the improvement in gas exchange is usually progressive, and it may take hours to appear; moreover, some of these patients show a progressive decrease in arterial oxygenation if the prone position is maintained for long periods of time. Thus, the long-term response to the prone position ranges from deterioration to great improvement, and it may be time dependent (slow improvement and slow deterioration).
3. The issue is further complicated when analyzing the effects of returning the patients to the supine position after they have been prone. Three different responses have been observed in patients who exhibited improved oxygenation in the prone position: (a) most of the patients display better oxygenation than in the original supine position, although the values are lower than in the prone position; (b) some patients return to basal supine oxygenation values when returned to the supine position (these patients are then

considered “prone dependent”); and (c) some patients display improved oxygenation compared to both the previous supine and prone values.

4. When the same patient is turned several times, the effects of the maneuver may change in time (no improvement, whereas previous positioning in the prone position produced marked improvement or the reverse).

From the above description, it is obvious that the effects of positioning depend on the individual patient (i.e., the underlying pathology and pathophysiologic status, which may vary over time). Over the years, several authors have described the response to prone position in specific categories of ARDS^{67,75} and patients without ARDS^{76,88} and the improvement in oxygenation induced by inhaled nitric oxide.^{61,62,64,65,70,74} The predictability of the oxygenation response to prone positioning, however, remains uncertain.⁷¹

Mechanisms Involved in Arterial Oxygenation Changes in the Prone Position

We can attempt to infer the mechanisms involved in arterial oxygenation changes when patients are turned to the prone position. We first discuss the mechanisms that may operate in the early period after positioning and then the mechanisms that are possibly time related.

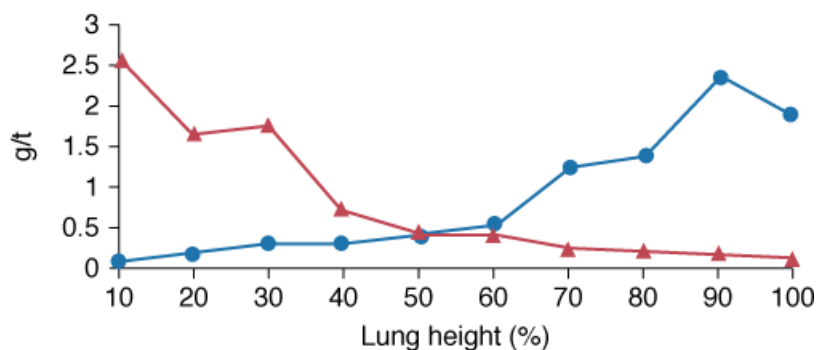
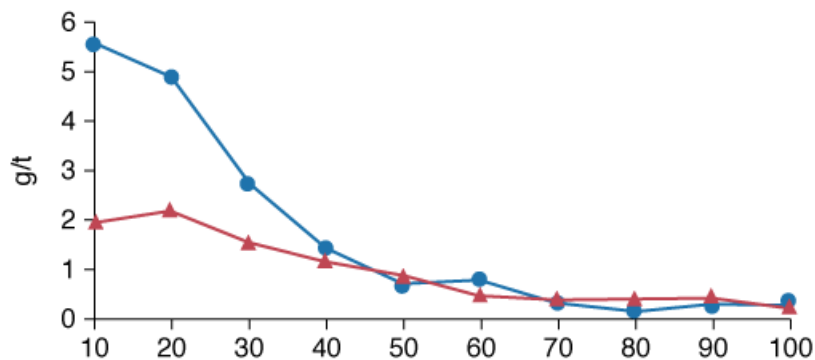
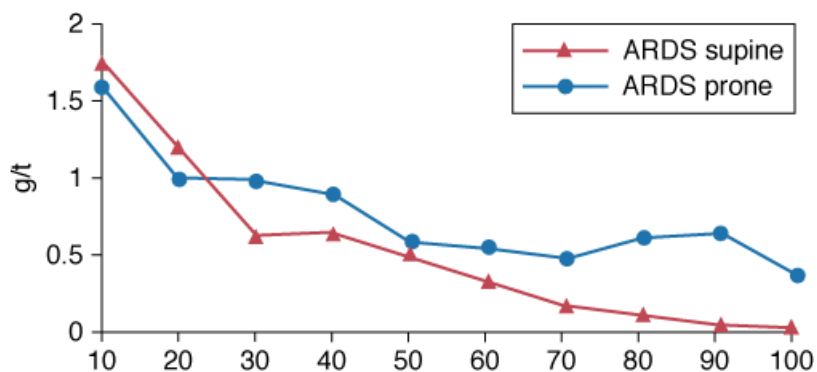
Early Effects

An increase or decrease in is caused by changes in Pa_{O_2} regional lung inflation, ventilation distribution, and perfusion distribution. The effects on oxygenation (improvement or deterioration) are likely explained by a varying interaction between the mentioned factors in the different postures.

As previously discussed in the section “[Regional Lung Inflation](#)” regional inflation decreases from ventral to dorsal in the supine position, and from dorsal to ventral in the prone position. The rate of decrease in regional inflation, however, is steeper in the supine position than in the prone position (see [Fig. 49-2](#)). Because the decrease in inflation may be expressed in exponential form, this is equivalent to saying that the decay rate of regional inflation along the vertical axis (K_d) is lower in the supine position than in the prone position. This, however, refers to average values. In an individual patient, the pattern may be substantially different.

In [Figure 49-5](#), we report three ARDS patients who showed different patterns of regional inflation in the supine and prone positions. In the first patient (*upper panel*), regional inflation behaves as for the average population (see [Fig. 49-2](#)). The exponential fitting of the decay of regional inflation ($r=0.91$) showed a K_d of 4.2 cm in the supine position. In the prone position, the decay of regional inflation was smoother, as reflected by the higher value of K_d , 14 cm (exponential fitting, $r=0.97$). In this patient, Pa_{O_2} improved by 65 mm Hg on moving from the supine to the prone position.

Figure 49-5



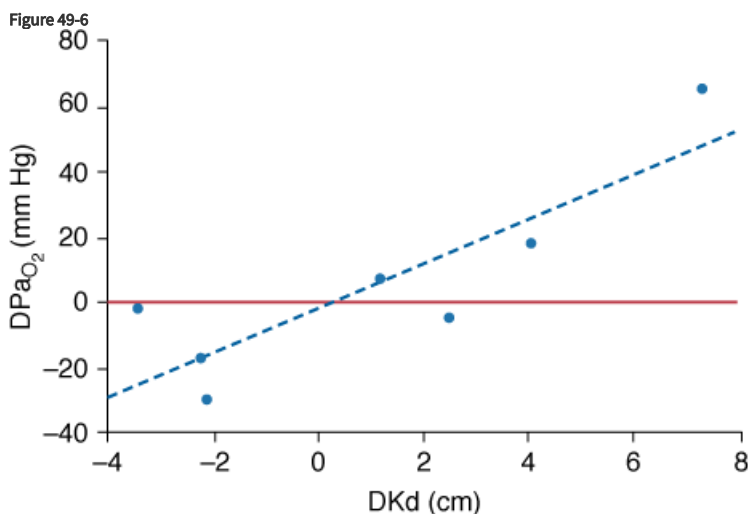
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Different behaviors of regional inflation (gas-to-tissue ratio [g/t]) as a function of lung height in the supine (*red triangles*) and prone (*blue circles*) positions in three acute respiratory distress syndrome (ARDS) patients. Zero on the height scale refers to the ventral lung surface in the supine position and to the dorsal lung surface in the prone position. As such, the direction of the lung scale is always from the nondependent to dependent lung region (see text for further explanation).

The *middle panel* in Figure 49-5 shows the regional inflation pattern of a patient in which the decay of regional inflation was significant in both the supine and prone positions ($r = 0.96$ and $r = 0.98$, respectively). In this case, however, K_d was lower in the prone position than in the supine position (4.3 and 9.4 cm, respectively). In this case, Pa_{O_2} deteriorated when the patient was turned from supine to prone (101 mm Hg when supine and 71.5 mm Hg when prone).

The *bottom panel* in Figure 49-5 refers to a very unusual patient. In this case, the low regional inflation areas in the supine position did not redistribute on switching to the prone position, and the pattern of regional inflation remained the same in the two positions. In this case, Pa_{O_2} did not change with alteration in position (61 mm Hg when supine and 68 mm Hg when prone).

We then tested the hypothesis that, in the individual patient, improvement in PaO_2 on switching from supine to prone occurs when the decay in regional inflation is smoother (i.e., a higher K_d) in the prone position, whereas a deterioration in PaO_2 occurs on switching from the supine to the prone position when the decay in regional lung inflation is steeper in the prone position (lower K_d). Plotting the individual K_d differences between prone and supine positions (DK_d) and the individual changes in PaO_2 (DPaO_2), we found a significant correlation between DK_d and DPaO_2 (Fig. 49-6). This suggests that the response in PaO_2 is associated with a change in regional inflation: The position in which regional inflation is more homogeneous (higher K_d) is associated with a better PaO_2 . When the regional inflation gradient along the vertical axis does not differ between the prone and supine positions (i.e., equal decrease from nondependent to dependent lung regions, whatever position is assumed), the PaO_2 does not change. To state that the PaO_2 changes are caused by regional inflation changes and not simply associated with them, however, we must assume that (a) ventilation behaves like regional inflation (i.e., it is more homogeneously distributed when inflation is more homogeneously distributed), and (b) perfusion is not changed when the position is changed (i.e., the relative perfusion of the dorsal and ventral regions does not change with position).



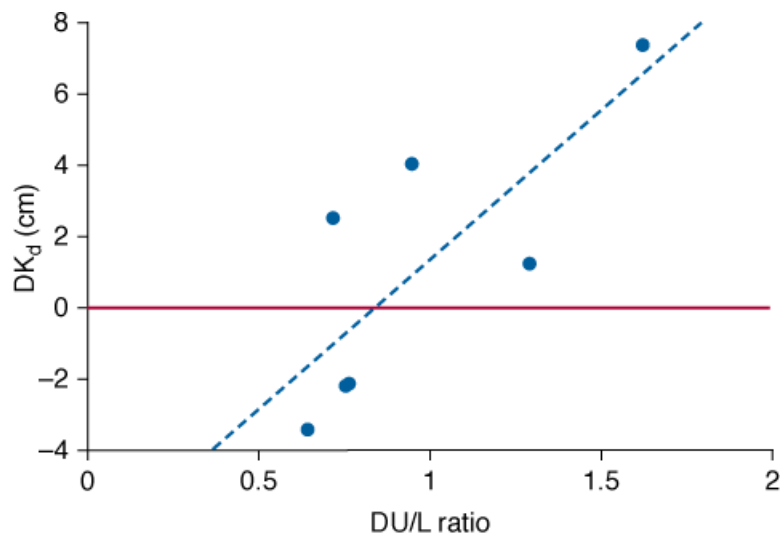
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Relationship between changes in PaO_2 (DPaO_2) and changes in the regional lung decay constants (DK_d) between the prone and supine positions. Data are from seven patients with ARF (regional inflation was computed as weighted mean of the two lungs). Positive values of DPaO_2 indicate that the PaO_2 is higher in the prone position, while negative PaO_2 values indicate that PaO_2 is higher in the supine position. When DK_d is close to zero (i.e., the decrease of regional inflation is similar in the prone and supine positions), the DPaO_2 is close to zero ($\text{D}\text{PaO}_2 = -1.7 + 6.6 \times \text{DK}_d$; $r = 0.86$; $P < .05$).

The question remains as to why in some patients regional inflation is more homogeneous in the prone position (and oxygenation improves in this position), while in other patients regional inflation is more homogeneous in the supine position (and oxygenation is better in this position). As previously discussed, regional inflation is dictated by Ppl , and this may, in turn, be influenced by SP_L and lung shape. Because SP_L is similar in the supine and prone positions,⁶ it is possible that the postural changes in regional inflation are caused by the associated lung-shape variations. We attempted to quantify the variation in lung shape on switching from the supine to the prone position by dividing the lung area into two compartments (at 50% of the ventral–dorsal distance), the upper (U) and the lower (L) compartments, and calculating the upper-to-lower (U/L) ratio in the prone and supine positions. The difference in U/L ratio between the prone and supine positions (DU/L ratio) gives a quantitative estimate of lung-shape change. For example, when the DU/L ratio is zero, lung shape is unmodified, and the greater the DU/L ratio, the greater the upper compartment is in the prone position. We found that the DU/L ratio significantly correlated with both DK_d (Fig. 49-7) and DPaO_2 (Fig. 49-8).

Figure 49-7

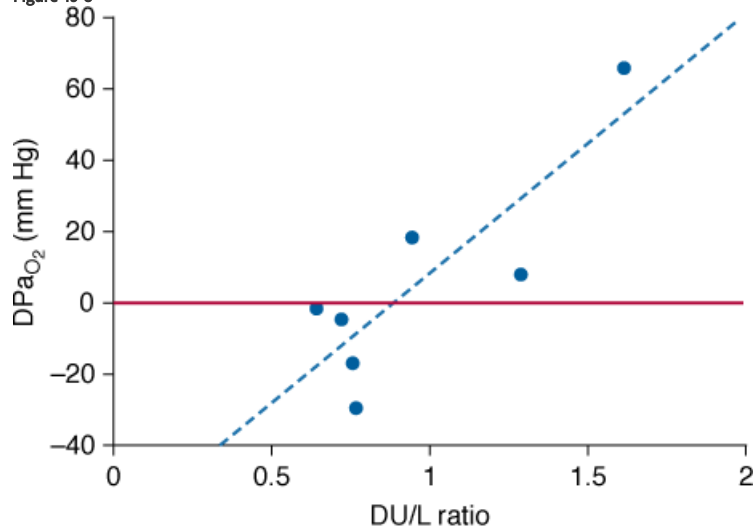


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Changes in the regional lung inflation decay constants (DK_d) as a function of the changes of upper-to-lower lung ratio (DU/L ratio) between the prone and supine positions. ($DK_d = -7 + 8.3 \times DU/L$ ratio; $r = 0.77$; $P < .05$).

Figure 49-8



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DPaO₂ as a function of the changes in upper-to-lower lung ratio (DU/L ratio) between the prone and supine positions ($DPaO_2 = -63 + 71.2 \times DU/L$ ratio; $r = 0.85$; $P < .05$).

This indicates that, when the U/L ratio is much lower in the supine position and much higher in the prone position, oxygenation will improve in the prone position. Consequently, we suggest that, according to these data, lung-shape modifications, as well as the modifications in regional inflation, may be predictive, in individual patients, of the early effects of positioning on arterial oxygenation.

Late Effects

When the positioning maneuvers are prolonged over time or are repeated several times during the course of ARF, the oxygenation response may be highly variable. No data are available with which to speculate on the possible mechanisms. We know, however, that with time edema may redistribute, lung pathology may progress from edema to fibrosis, and vascular characteristics may change. Further studies are needed to understand the time-related effects of positioning on arterial oxygenation.

Effects on Arterial CO₂

Recently there has been a paradigmatic shift of attention from changes in PaO_2 to changes in partial pressure of arterial carbon dioxide (PaCO_2) consequent to prone positioning in patients with ARF. According to PaCO_2 change, patients may be divided into two categories: those who show a decrease of PaCO_2 secondary to prone position, and those whose PaCO_2 does not change or even increases.

Before discussing the meaning of PaCO_2 change, it is worth remembering that expected changes in PaCO_2 are relatively smaller than changes in PaO_2 because of the different slopes of the content/tension relationship. For a similar change in content, the change in tension of CO_2 (partial pressure of the gas) is expected to be approximately 12% of the change in PaO_2 . This would result, for instance, in a change in PaO_2 of 5 mm Hg versus a change in PaO_2 of 40 mm Hg, implying not only more controlled measurements, but also greater accuracy.

That PaCO_2 increases during the course of ARDS, in association with structural changes of the lung, has been known for many years.^{77,78} Kallet et al found that an increase in dead space is a prognostic marker of ARDS mortality.⁸⁹

We found that patients who display a decrease in PaCO_2 in the prone position experience a greater survival (dose dependent) than do patients who do not show a decrease, or even show an increase in PaCO_2 .⁹⁰ Moreover, we recently reported the decrease in PaCO_2 during prone ventilation is associated with a higher lung recruitability as measured with CT scan.⁹¹ Accordingly, the PaCO_2 response to prone positioning may be associated to the underlying pathology: predominant recruitment in PaCO_2 responders versus blood-flow diversion in nonresponders. Interestingly, only the changes in PaCO_2 were associated with survival in this study, not the changes in PaO_2 .

The Impact of Prone Positioning on Mortality

Rationale

Since the first report dating back to the 1970s,¹⁻³ the prone position has been proposed and used to improve oxygenation. The progressive recognition of the mechanism underlying this effect, however, led to the hypothesis that the prone position may decrease the danger associated with mechanical ventilation (ventilator-induced lung injury).⁹² The hypothesis is based primarily on the observation that in the prone position there is a more homogeneous distribution of inflation, suggesting indirectly a more homogeneous distribution of stress and strain, which are likely the first triggers of ventilator-induced lung injury. In fact, in several animal models, it has been shown that the prone position decreases ventilator-induced lung injury or at least delays its progression.^{9,93,94} CT scan studies in animals documented a decrease in regional strain in the prone as compared with the supine position.⁹ Indeed, the possible favorable effects of prone positioning on outcome of patients with ARF should not be related to oxygenation changes, which were not found to be associated with outcome, but, instead, to changes in regional lung mechanics, by which forces applied by the ventilator should be more evenly distributed throughout the lung parenchyma.

Randomized Clinical Trials on Prone Positioning

To date, four major randomized, controlled trials (RCTs) have addressed the effect of prone positioning on outcome of adult patients with hypoxemic ARF.⁷⁷⁻⁸⁰ Other minor published trials will not be discussed here, because they were characterized by major limitations (e.g., small sample size, inclusion criteria other than adult hypoxemic respiratory failure, application of prone positioning for less than 48 hours, patients cotreated with nonconventional ventilator modalities, and so on).^{76,81,95-100} Table 49-1 lists the main characteristics of the four largest published trials. As shown, these trials exhibit large variations in term of enrollment criteria (acute lung injury or ARDS vs. hypoxemic respiratory failure vs. ARDS only), intensity of intervention ("early phase strategy" in the trials published earlier vs. "prolonged strategy" in the latest trials), and use of relevant cointervention (i.e., protective mechanical ventilation strategy). Despite the incorporation of the successively acquired scientific evidence into the newer protocols, none of these trials showed any benefit of prone positioning on the mortality of the general population of patients with acute lung injury or ARDS. In a post hoc analysis, however, of one of the first RCTs evaluating prone positioning,⁷⁷ we reported a reduction in 10-day mortality in the subgroup of patients with severe

hypoxemia or with higher severity of illness. This finding was reinforced in the most recent RCT,⁸⁰ where the subgroup of patients with the most severe ARDS (stratified at randomization according a $\text{Pa}_{\text{O}_2}/\text{Fi}_{\text{O}_2}$ below 100) exhibited a trend toward a reduction in absolute 6-month mortality in the treatment arm of approximately 10%, a signal that did not reach statistical significance because of the small number of patients included in this subgroup. Notably, patients treated with prolonged prone ventilation demonstrated an increase in the rate of adverse events in this study (e.g., dislodgment of chest tube or endotracheal tube, desaturation, need for increased sedation and muscle relaxants, hypotension, or arrhythmias).

Table 49-1: Characteristics of the Largest Randomized, Controlled Trials Investigating the Effect of Prone Positioning on the Outcome of Patients with Hypoxemic Acute Respiratory Failure

	Taccone et al ⁸⁰	Mancebo et al ⁷⁹	Guérin et al ⁷⁸	Gattinoni et al ⁷⁷
Patients (n)	344	142	802	304
Enrollment period (years)	2004 to 2008	1998 to 2002	1998 to 2002	1996 to 1999
Enrollment criteria	ARDS with PEEP ≥ 5 cm H ₂ O	ARDS with four-quadrant infiltrates on CXR	Hypoxemic acute respiratory failure (413 ALI/ARDS pts)	ALI/ARDS with PEEP ≥ 5 cm H ₂ O
Stratified randomization for ARDS severity	Yes	No	No	No
Time after meeting enrollment criteria	<72 hours	<48 hours	>12 to 24 hours	Not prespecified
Last follow-up	6 months	Hospital discharge	90 days	6 months
Planned duration of prone positioning (average)	20 hours/day for 28 days	20 hours/day until weaning criteria	≥ 8 hours/day until weaning criteria	6 hours/day for 10 days
Actual duration of prone positioning (average)	18 hours for 8.3 days	17 hours for 10.1 days	9 h for 4.1 days	7 hours for 4.7 days
Protective mechanical ventilation	Yes ($V_T \leq 8$ mL/kg of PBW)	Yes ($V_T \leq 10$ mL/kg of PBW or ABW)	No	No
Trial ended early	No	No	Yes (slow enrollment)	Yes (slow enrollment)

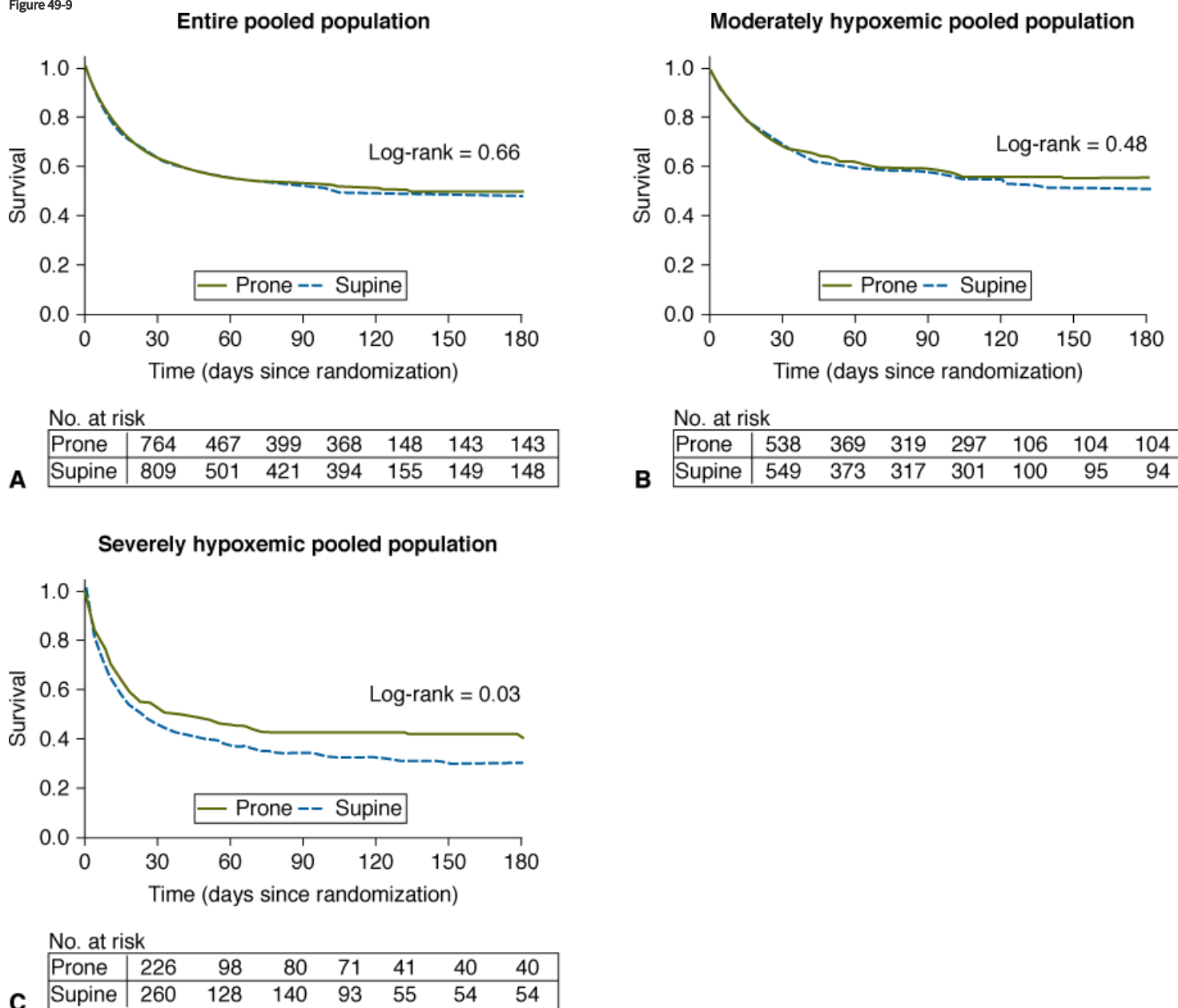
Abbreviations: ABW, actual body weight; ALI, acute lung injury; ARDS, acute respiratory distress syndrome; CXR, chest X-ray; PBW, predicted body weight; PEEP, positive end-expiratory pressure; V_T , tidal volume.

Meta-Analyses on Prone Positioning

A major limitation of the reported RCTs on prone positioning is the relatively small sample size, which could result in an underpowered study, particularly in the case of subgroup analysis. Because of the low incidence of ARDS, especially in its most severe form, several years of data collection are required, which prolongs the time needed to conduct an RCT on prone positioning.¹⁰¹ Consequently, several meta-analyses have been performed to evaluate the effect of prone positioning in a larger population of patients.^{82,84–86,102} Of note, most of these meta-analyses were conducted on a study level, reporting inconclusive results on primary and secondary outcomes in the

unselected population of patients with hypoxemic ARF. This kind of meta-analysis, however, does not permit any subgroup stratification, whereas the body of knowledge derived from RCTs clearly suggests that prone positioning could be beneficial only in the subset of the most severely ill patients. To overcome this issue, two meta-analyses^{83,103} with different trial inclusion criteria, have been conducted at a patient level, in collaboration with prone trialists, to determine whether prone positioning has a different impact on mortality depending on the extent of hypoxemia. In both of these studies, a threshold of PaO_2 -to- FI_{O_2} ratio of 100 mm Hg was chosen, in line with the most recently published RCT.⁸⁰ Both meta-analyses support the hypothesis that prone positioning decreases mortality of patients with severe hypoxemia, whereas it has no beneficial effect in patients with less-severe hypoxemia (Fig. 49-9). Interestingly, a post hoc analysis using varying PaO_2 -to- FI_{O_2} thresholds suggested improved mortality in the most severely hypoxemic group, using a PaO_2 -to- FI_{O_2} ratio of up to 140 mm Hg to define this subgroup.⁸³

Figure 49-9



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Kaplan-Meier estimates of survival rates at the last follow-up of the prone (continuous green line) and supine (dotted blue line) patients from a patient-level meta-analysis of the four largest RCTs investigating the effects of prone positioning on mortality:⁷⁷⁻⁸⁰ (A) entire ARF

population; (B) moderately hypoxemic patients ($\text{PaO}_2/\text{FiO}_2$ 100 to 200 mm Hg at baseline; and (C) severely hypoxemic patients (i.e., $\text{PaO}_2/\text{FiO}_2 > 100$ mm Hg). (Used, with permission, from Gattinoni et al.¹⁰³)

We speculate that patients with severely hypoxemic ARDS derive benefit from prone positioning because it is only this subgroup of patients who have the physiologic prerequisites for prone positioning to work: They have a greater amount of edema, more widespread alveolar collapse, and higher lung recruitability.¹⁰⁴ In fact, the benefit of prone positioning on the respiratory system is greater recruitment of the dorsal region, which could lead a more homogeneous distribution of the stress and strain forces of mechanical ventilation, potentially decreasing the risk of alveolar injury by mechanical ventilation. This phenomenon is best observed when the aerated portion of the lung is small and when lung recruitability is high.¹⁰⁵

Summary

The evidence derived from RCTs and recent meta-analyses clearly indicate that prone positioning should not be implemented in patients with less-severe hypoxemic respiratory failure, a situation in which therapy produces useless and potentially harmful effects. Prone positioning, however, should be considered for the treatment of patients with severe hypoxemia, although great care and experienced personnel are required when performing this intervention. Future studies are needed to clarify practical questions facing the clinicians using prone positioning, such as the optimal duration of the therapy,⁸⁵ the validity of the PaO_2 -to- FiO_2 threshold⁸³ in guiding therapy, and the real impact of adverse events. A currently ongoing trial targeting the enrollment of 460 patients with $\text{PaO}_2/\text{FiO}_2$ less than 150 mm Hg will provide additional data.¹⁰⁶

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